All The Mass We Cannot See: Verifying The Presence of Dark Matter In The Milky Way

Riya Kore, Leah Napiwocki, Ritvik Sai Narayan, Alex Tellez, Nina Weichmann

Advisors: Prof. Snezana Stanimirovic, Dr. Ralf Kotulla

Department of Astronomy, University of Wisconsin-Madison

Introduction/Motivation

The purpose of this project is to measure the rotation curve of the Milky Way galaxy. Several key pieces of information can be derived from this measurement, one related to understanding dark matter. Assuming that stars and neutral hydrogen (HI) clouds in the Milky Way move along circular orbits around the Galactic Center, the radial velocity component of star/cloud orbital motion (V_r) is expressed as:

$$V_r = R_* \sin l \left(\frac{V}{R} - \frac{V_*}{R_*}\right)$$

In this equation l is the Galactic longitude of the observed star/cloud, V and V_*(sun) are the orbital velocities of the star/cloud and the Sun, and R and R_*(sun) are the distances from GC to the star/cloud and the Sun. By knowing the Galactic rotation curve(V_r), the equation can be used to approximate the radial velocity expected to measure in different regions of the Milky Way. By plotting V_r as a function of Galactic longitude a diagram of $l - V_r$ is obtained. Knowing the Galactic rotation curve and measuring V_r , the Galactocentric distance R to the particular star/cloud is also estimated. This value is directly related to the distance from the Sun to the particular star/cloud. These distances are called "kinematic distances" because they depend on the assumed rotation curve.

Another value that can be approximated is the total mass of the Milky Way (M_r) enclosed within the specified Galactocentric radius R using this relation:

$$M_r = \frac{V^2 R}{G}$$

with G being the gravitational constant. The shape of the rotation curve gives insight on the distribution of mass in the Milky Way. The Milky Way rotation curve stays flat until about $R = 10kpc$ with indication that it increases beyond this radius. Due to Newtonian mechanics if most of the mass is distributed within a particular radius, than beyond that radius the expected orbital velocity is $\propto R^{1/2}$. Because observational data does not support this in the case of the Milky Way and most other spiral galaxies, this suggests the existence of dark matter that extends beyond visible galaxy disks.

Based on geometric considerations there is a set of points in the Milky Way disk where the distance can be estimated. These points are tangent points and the method for measuring the Galactic rotation curve is the Tangent-point Method. Tangent points are located in the inner Galaxy and are the closest to GC for any longitude direction and are located in the Galactic plane on the circle passing through GC and the Sun and having a diameter of R_*. For each tangent point, the geometry is simple and results in:

$$R = R_* \sin l$$

$$V = V_r + V_* \sin l$$

From these two relations V(R) can be estimated. Tangent points exist only for 0< l <90 degrees and 270 < l < 360. The expected V_r for 75 < l < 90 degrees and 270 < l < 295 degrees is very small and close to 0 km/s which makes it best to avoid those longitudes due to large velocity crowding. The longitudes close to the Galactic center(l < 20) have highly non-circular orbital motions because the density is very high and other dynamical features like the stellar bar exist. Therefore, it is best to collect observations from the 20 to 75 degree range and observe every FWHM.

Observations and Methods

We used the 2.3 meter Pine Bluff Small Radio Telescope for our Observations. We set the telescope to a central frequency of 1420.4 MHz, which is the frequency of the 21-cm emission line of atomic hydrogen. Then, we got out data following these steps:

- Performed noise calibration going from galactic latitude 30 to 0.
- After reaching galactic latitude 0, we collected data along the galactic plane going from galactic longitude 20 to 74 incrementing it by 6 degrees everytime.
- At each position, we had an integration time of 300 seconds.

The next step was converting the data from frequency to LSR radial velocity for which we used a couple of steps:

- Averaged the collected intensity (in units of Antenna Temperature [K]) for each offset to get the combined integrated intensity for each galactic longitude.

We performed this averaging to reduce statistical noise and ensure a more reliable signal for our subsequent analyses.

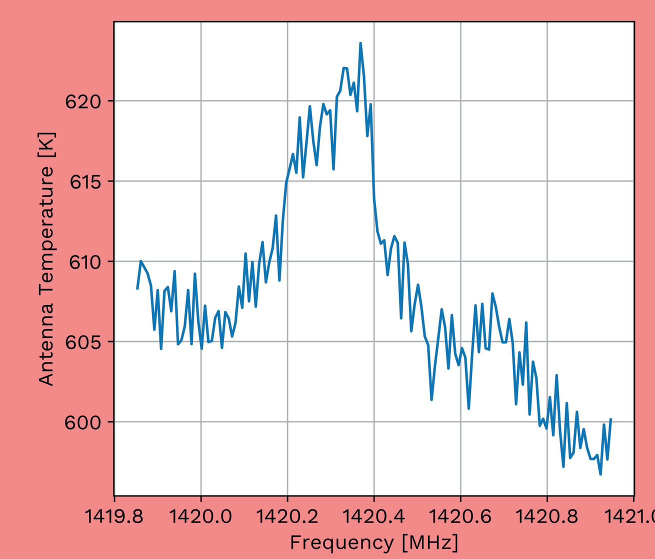

Figure 1: Averaged Intensity of HI spectrum (in units of Antenna Temperature) vs. Frequency (in MHz) for the offset 54.

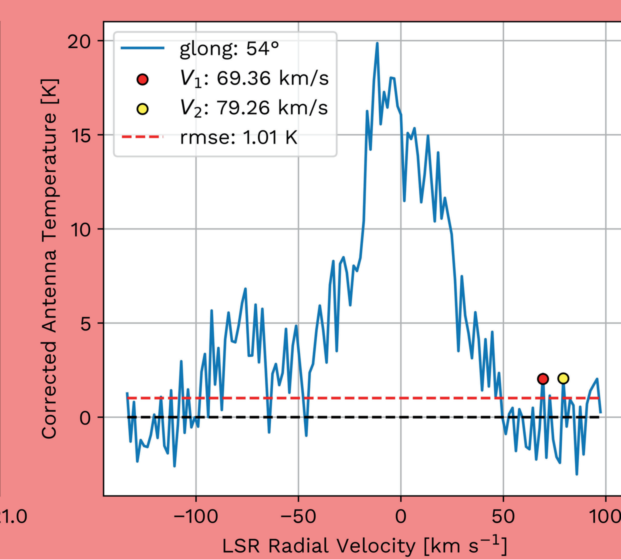

Figure 2: Baseline subtracted Average Intensity of HI spectrum (in units of Antenna Temperature) vs. LSR Radial Velocity (in km/s) for the offset 54 with the highest positive radial velocity marked in red and yellow.

- Converted the Frequency to Topocentric Radial Velocity (in km/s) using the doppler shift equation.

$$\frac{V_r}{c} = \frac{f_0 - f}{f_0}$$

- Subtracted the spectral baseline we got from noise calibration
- Converted the Topocentric Radial Velocity to Local Standard of Rest (LSR) Radial Velocity to compare with published results.

- Estimated two tangent point radial velocities:
  - V_1: Largest velocity above one sigma
  - V_2: Largest velocity corresponding to a clear peak in the spectrum
- V_1 and V_2 were averaged to get the Radial Velocity V_r and 12.3 km/s was subtracted to account for the random motions of the HI clouds along LOS.
- Calculated the Circular Velocity from the Radial Velocity for each offset.

Results - Rotation Curve

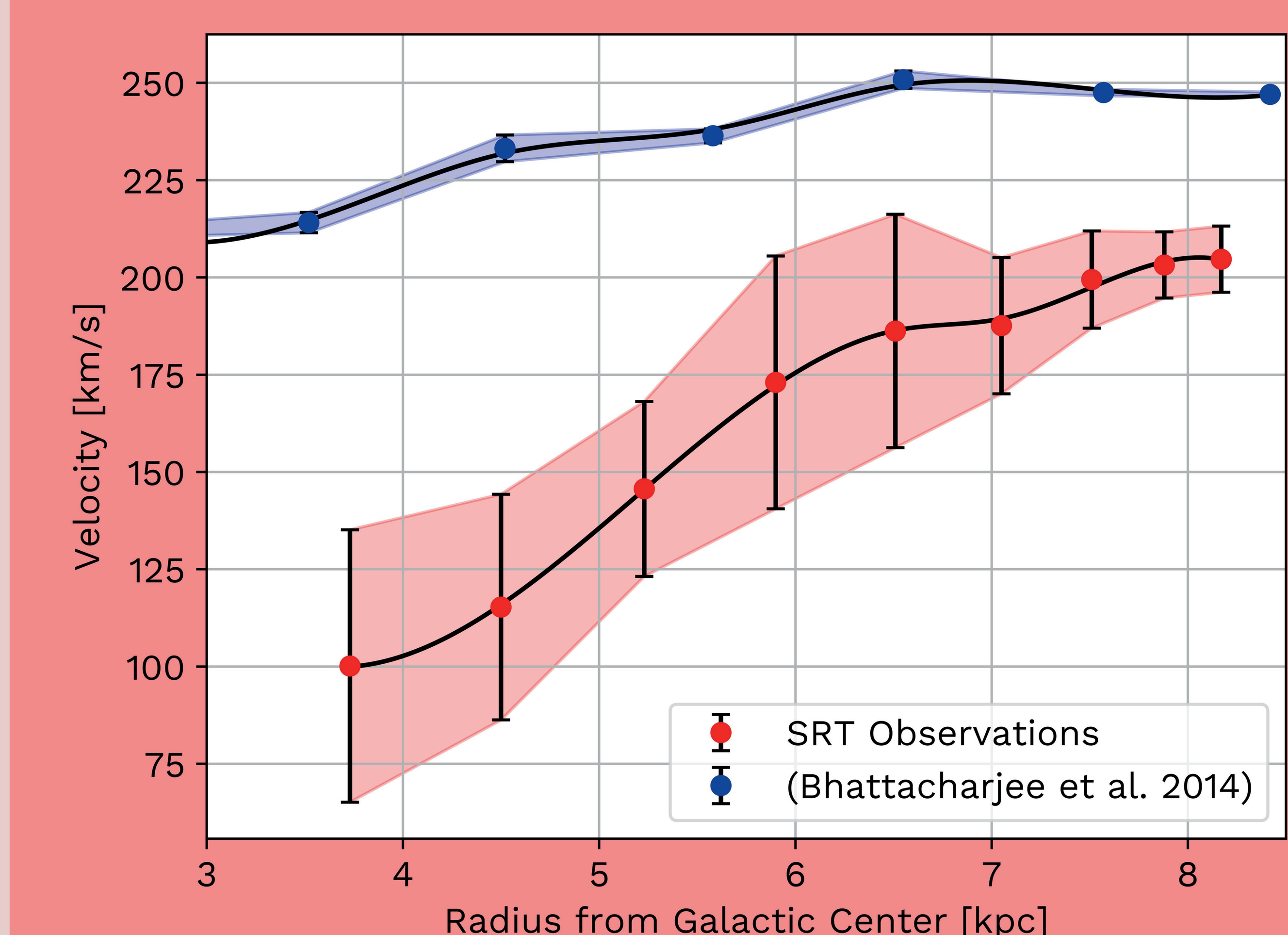

Figure 3: Galactic Rotation Curve - the velocity of hydrogen clouds in km/s versus their distance from the Galactic Center in kpc. This work's observations are shown in red; observations that have been published by Bhattacharjee et al. are shown in blue.

The differences our results compared to those of Bhattacharjee et al. can be explained by the quality of the SRT and noise. Despite this, we observe the anticipated trend: the velocity increases with radius before leveling out.

Results - Enclosed Mass

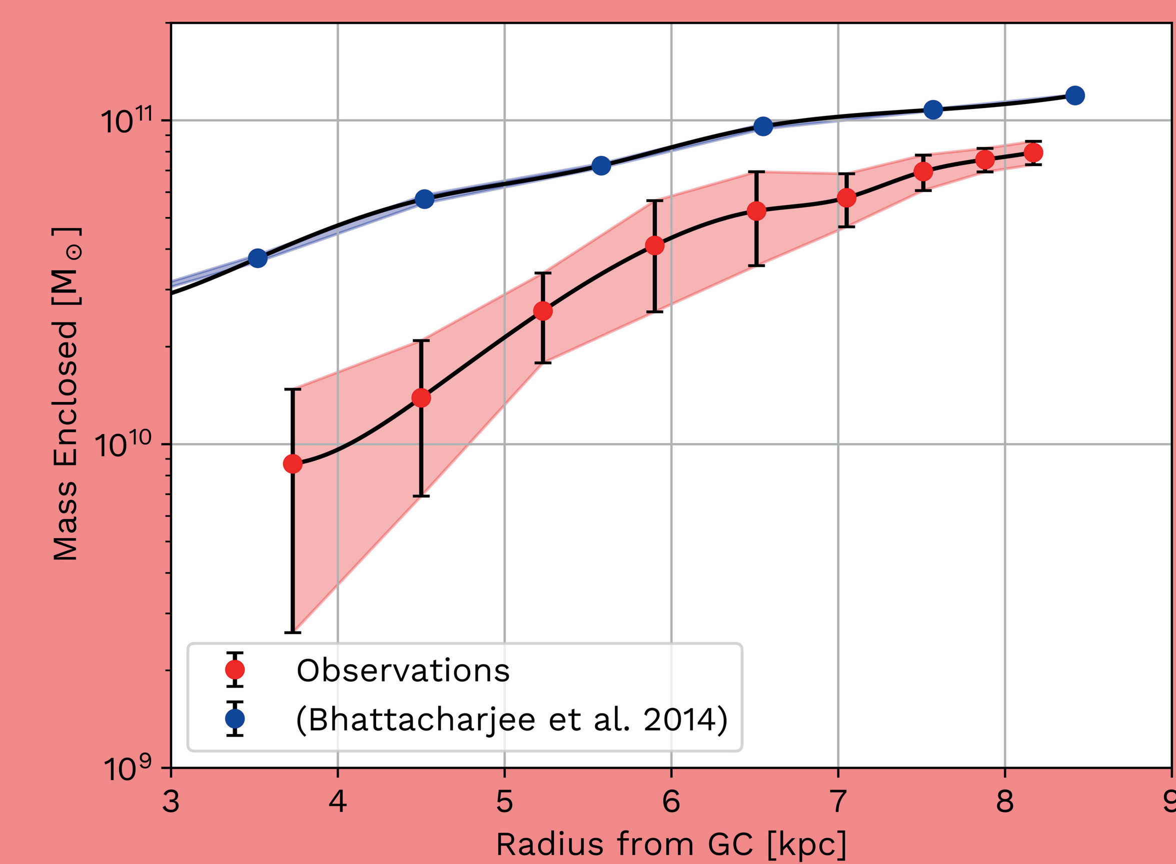

Figure 4: Enclosed Mass Function - the mass of matter within the Milky Way in solar masses as a function of distance from the Galactic Center in kpc. This work's observations are shown in red; observations that have been published by Bhattacharjee et al. are shown in blue.

Once again, our observations gave lower values than anticipated; however, the trend of the mass increasing with distance is clearly seen. It is known that using optical telescopes, the radius of matter in the Milky Way is smaller than what is observed using radio telescopes. This implies the existence of dark matter that cannot be seen in optical due to it not reflecting light. Our results agree with this implicaiton.

Key Takeaways

- Observed Rotation Curve
  - Based on neutral hydrogen gas clouds, the rotation curve remains **flat,** even at large distances from the galactic center (figure 3)
  - Theoretical models predict that radial velocities should **decrease** with distance, but this is not observed
- Mass Distribution
  - The enclosed mass of the galaxy **increases continuously** with distance from the center (figure 4)
  - Theoretical observations suggest a sudden **drop-off** in mass at the galaxy's edge, but no such cutoff is observed
- Dark Matter Evidence
  - The additional mass needed to explain these observations cannot be attributed to stars, gas, or dust
  - The only explanation is the presence of **dark matter-** a mysterious form of matter that does not absorb/emit light, but interacts gravitationally
- Implications
  - The flat rotation curve and the increasing enclosed mass suggest the presence of a massive, extended dark matter halo
  - This halo dominates the galaxy's gravitational potential and is essential to explain the observed orbital velocities at large radii

References

- Pijushpani Bhattacharjee *et al* 2014 *ApJ* 785 63
- Sparke, Linda S, and John S. Gallagher. Galaxies in the Universe: An Introduction. Cambridge Univ. Press, 2010
- Stanimirovic, Snezana. "Rotation Curve." Astronomy 465. Astronomy 465, Oct. 2024